

Design Thinking and Innovation

SITAMS CSE (AI & ML) • R23 • II Year • IV Semester • Course Code: 23ESC241I

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: INTRODUCTION TO DESIGN THINKING Introduction to elements and principles of Design, basi...

Syllabus topics

- INTRODUCTION TO DESIGN THINKING Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components
- Principles of design
- Introduction to design thinking, history of Design Thinking, New materials in Industry

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit II: DESIGN THINKING PROCESS Design thinking process (empathize, analyze, idea & prototype),...

Syllabus topics

- DESIGN THINKING PROCESS Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations
- Tools of design thinking - person, costumer, journey map, brainstorming, product development Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc
- Every student should explain about product development

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Unit III: INNOVATION Art of innovation, Difference between innovation and creativity, role of cre...

Syllabus topics

- INNOVATION Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations- Creativity to Innovation- Teams for innovation- Measuring the impact and value of creativity
- Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation

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Unit IV: PRODUCT DESIGN Problem formation, introduction to product design, Product strategies, P...

Syllabus topics

- PRODUCT DESIGN Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications- Innovation towards product design- Case studies Activity: Importance of modelling, how to set specifications, Explaining their own product design

Revision checklist

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Unit V: DESIGN THINKING IN BUSINESS PROCESSES Design Thinking applied in Business & Strategic I...

Syllabus topics

- DESIGN THINKING IN BUSINESS PROCESSES Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining
- Relevance, Extreme competition, Standardization
- Design thinking to meet corporate needs- Design thinking for Startups- Defining and testing Business Models and Business Cases- Developing & testing prototypes
- Activity: How to market our own product, About maintenance, Reliability and plan for startup

Revision checklist

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This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.